

University of Nottingham Ningbo

Research Ethics Checklist for Staff and Research Students

[strongly informed by the ESRC (2012) *Framework for Research Ethics*]

A checklist should be completed for **every** research project or thesis where the research involves the **participation of people, the use of secondary datasets or archives relating to people and/or access to field sites or animals**. It will be used to identify whether a full application for ethics approval needs to be submitted.

You must not begin data collection or approach potential research participants until you have completed this form, received ethical clearance, and submitted this form for retention with the appropriate administrative staff.

The principal investigator or, where the principal investigator is a student, the supervisor, is responsible for exercising appropriate professional judgement in this review.

Completing the form includes providing brief details about yourself and the research in Sections 1 and 2 and ticking some boxes in Sections 3 and/or 4, 5, 6.

Ticking a shaded box in Sections 3, 4, 5 or 6 requires further action by the researcher. Two things need to be stressed:

- Ticking one or more shaded boxes does **not** mean that you cannot conduct your research as currently anticipated; however, it does mean that further questions will need to be asked and addressed, further discussions will need to take place, and alternatives may need to be considered or additional actions undertaken.
- Avoiding the shaded boxes does **not** mean that ethical considerations can subsequently be 'forgotten'; on the contrary, research ethics - for everyone and in every project - should involve an ongoing process of reflection and debate.

The following checklist is a starting point for an ongoing process of reflection about the ethical issues concerning your study.

SECTION 1: THE RESEARCHER(S)

1.1: Name of principal researcher: [REDACTED]

1.2: Status: ☐ Staff
☒ Postgraduate research student

1.3: School/Division: School of Computer Science

1.4: Email address: [REDACTED]

1.5: Names of other project members (if applicable):

1.6: Names of Supervisors (if applicable): [REDACTED]
[REDACTED]

	Yes	No
1.7: I have read the University of Nottingham's <i>Code of Research Conduct and Research Ethics</i> (2010) and agree to abide by it: http://www.nottingham.edu.cn/en/research/researchethics/ethics-approval-process.aspx	☒	☐
1.8: (If applicable) I have read the University of Nottingham's <i>e-Ethics@Nottingham: Ethical Issues in Digitally Based Research</i> (2012) and agree to abide by it. http://www.nottingham.edu.cn/en/research/documents/e-ethics-at-the-university-of-nottingham.pdf	☒	☐
1.9: When conducting research on people (Section 5) I will prepare both a <i>participant consent form</i> as well as a <i>participant information sheet</i> . I am aware that the following templates are available on the Ethics webpage: http://www.nottingham.edu.cn/en/research/researchethics/ethics-approval-process.aspx • Participant consent form • Participant Information Sheet English and Chinese	☒	☐

SECTION 2: THE RESEARCH

2.1: Title of project:

Evaluating the Performance of a Sensory Management Recommendation System with Children with Autism Spectrum Disorders

Please provide brief details (50-150 words) about your proposed research, as indicated in each section

2.2: Research question(s) or aim(s)

Unusual sensory experiences are commonly seen in children with Autism Spectrum Disorder (ASD). These experiences can cause distress and affect many aspects of life in their lifetime. In China, ASD families are facing a number of cultural, financial and structural barriers. Innovative technology-based interventions that are easily accessible and affordable are needed by ASD families in China.

As part of my PhD program, I have designed and developed a mobile application which can measure the physiological and environmental data and provide real-time suggestions to help improve the comfort and attention of ASD children via sensory management. The developed application is called sensory management recommendation system (SMRS). Previously in 2019, the researcher has obtained the ethics panel's approval to the prototype testing of the SMRS. Further to test and evaluate the fully-developed SMRS, the researcher is applying for ethics panel's approval to this study.

The first goal of this study is to evaluate how accurately the SMRS identifies the abnormal attention and stress levels of the children with ASD in real-life cases of different conditions of sound, light and temperature. The second goal is to investigate the effectiveness of management strategies recommended by the SMRS on children attention improvement. Furthermore, the caregivers' level of satisfaction in terms of system utilisation, such as user interface, intention of long-term use, will be assessed as well.

2.3: Summary of method(s) of data collection

Subject Selection

The study will be performed on preschool-age children formally diagnosed with ASD and Typically Developed (TD) children. The subjects will be recruited from collaborative institutes, including a special education school and a local kindergarten. The study intends to recruit at least 60 children in total, categorised by two groups: the ASD group, consisting of at least 30 children diagnosed with ASD; and the TD group, consisting of at least 30 gender- and age-matched TD children. The Appendix I – Research Protocol gives more details about the inclusion and exclusion criteria for subject selection.

Data Collection

The data collection will be conducted through two stages. The first stage is a pilot testing under controlled environment to initially test the system performance. The second stage will evaluate and validate the performance and effectiveness of SMRS with ASD and TD subjects in real-life environment.

Stage 1 – Pilot Testing of SMRS in Controlled Environment

The pilot testing will only assess the feasibility and functional performance of the SMRS. The testing will be conducted in one classroom in the local kindergarten with a TD child. Parents of the participant should register for an account on the application and complete the Sensory Profile Questionnaire before the testing starts. The participant will be required to wear the sensing devices during the testing as shown in Figure 1. Fundamental functions such as sensor connection, network connection and data transmission will be tested. The controlled environment will then be created by adjusting different environmental conditions (i.e., temperature, noise, light intensity). One of the variables will be adjusted to levels from low to high and the other two variables should be controlled to be 'moderate' (see Table 1). The stimuli will last over 30 seconds, and investigator will check if the SMRS responds correctly in terms of data visualisation, prediction, and strategy making in the test cases. Further, the investigator will test different combination of sensory stimuli and check the system response outcomes.



Figure 1: A child wearing the sensing devices of the SMRS system

Table 1: The value of the stimuli

Variable	Values (from low to high)	Unit
Temperature	22, 24, 26, 28, 30	°C
Noise	40, 50, 60, 70, 80	dB
Light intensity	275, 325, 375, 425, 475	lx

Stage 2 – Testing of SMRS in Real-Life Environment

After the pilot testing, SMRS will be evaluated in real-life environment with ASD and TD groups along with their caregivers and teachers. The real-life environment refers to normal and regular environment in daily life in which there is no control on the generation of sound, visual and tactile related stimuli. Five small classrooms as shown in Figure 3 in Section 2.4 will be used as the testing sites for the ASD group. Another five classrooms with similar settings in a local kindergarten will be used as the testing sites for the TD group.

The data collection at this stage will use alternating no-intervention and intervention design. Each participant will go through three single-subject sessions: no-SMRS session, SMRS session 1 and SMRS session 2. In each session, one child will take regular class with one teacher for 30 minutes, whereas the caregiver should observe the whole session outside the room by remote monitor or through the window directly. Only in SMRS sessions will the subject be required to wear the sensing devices (i.e., Apple Watch and GSR sensor). Five testing rooms will be used to perform testing of SMRS at the same time as shown in Figure 2.

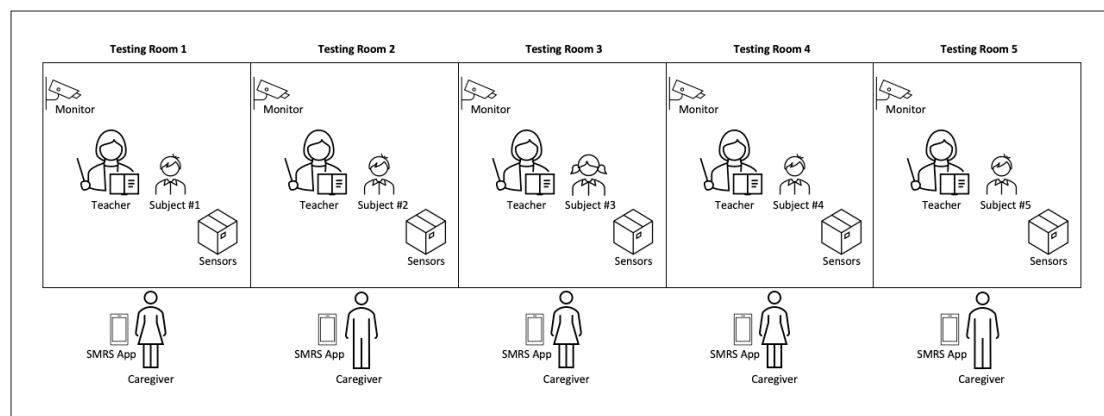


Figure 2: The Set-Up for a SMRS session

The no-SMRS session is for collecting baseline data. The subject will take class without SMRS as normal. In a SMRS session, when the SMRS identifies abnormal situation and then the system generates a recommended strategy, the teacher and caregiver will simultaneously receive a message of the recommended strategy. In SMRS session 1, the caregiver or teacher will not follow immediate strategies recommended by the system for attracting the child's attention or calming down the child. However, in SMRS session 2, the teacher should take actions within 30 seconds as instructed by the strategy to help the child pay attention or calm down during the class. After each session, the teacher and caregiver will be required to use validated measure tools to rate the child's performance and system accuracy. The Appendix I – Research Protocol has provided more details about the study design and measures.

2.4: Proposed site(s) of data collection

The data collection will be conducted in a special education school and a kindergarten in [REDACTED]. [REDACTED] special education school is a rehabilitation centre which provides training courses for pre-school children with ASD. [REDACTED] is a public kindergarten accepting preschool kids above age of three. The classroom as shown in Figure 3 will be used as the testing room. The room is equipped with air-conditioning, desk, chairs, play area and surveillance. Through the window on the door, parents and investigator can also observe the testing session from the outside.

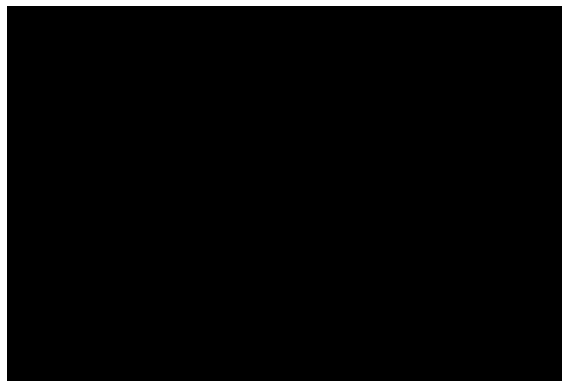


Figure 3: A Testing Room

2.5: How will access to participants and/or sites be gained?

The investigator has established networks with service providers and parental support groups within Zhejiang. The collaborative schools mentioned above have known the research project and agreed to provide sites and support for the testing.

Participants in this study will be recruited via purposive sampling strategy. [REDACTED] special education school will make contacts with parents of ASD children and recruit a group of ASD participants. [REDACTED] will recruit TD participants. Their contact information is enclosed in the last page of this document. The investigator will inform parents about the details of the project. Participant information sheet will be provided, and the informed consent will be obtained before the testing.

Professional teachers will be on-site to support the testing and ensure the ethical practices during the whole testing.

SECTION 3: RESEARCH INVOLVING USE OF SECONDARY DATASETS OR ARCHIVES RELATING TO PEOPLE

If your research involves use of secondary datasets or archives relating to people all questions in Section 3 **must** be answered. If it does not, please tick the 'not relevant' box and go to Section 4.

NOT RELEVANT	☐
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Please answer each question by ticking the appropriate box.

	Yes	No
3.1: Is the risk of disclosure of the identity of individuals low or non-existent in the use of this secondary data or archive?	☒	☐
3.2: Have you complied with the data access requirements of the supplier (where relevant), including any provisions relating to presumed consent and potential risk of disclosure of sensitive information?	☒	☐

SECTION 4: RESEARCH INVOLVING ACCESS TO FIELD SITES AND ANIMALS

If your research involves access to field sites and/or animals all questions in Section 4 **must** be answered. If it does not, please tick the 'not relevant' box and go to Section 5.

NOT RELEVANT	☒
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Please answer each question by ticking the appropriate box.

	Yes	No
4.1: Has access been granted to the site?	☐	☐
4.2: Does the site have an official protective designation of any kind?	☐	☐
If yes, have the user guidelines of the body managing the site a) been accessed? b) been integrated into the research methodology?	☐	☐
	☐	☐
4.3: Will this research place the site, its associated wildlife and other people using the site at any greater physical risks than are experienced during normal site usage?	☐	☐
4.4: Will this research involve the collection of any materials from the site?	☐	☐



4.5: Will this research expose the researcher(s) to any significant risk of physical or emotional harm?	☐	☐
4.6: Will the research involve vertebrate animals (fish, birds, reptiles, amphibians, mammals) or the common octopus (<i>Octopus vulgaris</i>) in any capacity?	☐	☐
If yes, will the research with vertebrates or octopi involve handling or interfering with the animal in any way or involve any activity that may cause pain, suffering, distress or lasting harm to the animal?	☐	☐

SECTION 5: RESEARCH INVOLVING THE PARTICIPATION OF PEOPLE

If your research involves the participation of people all questions in Section 4 **must** be answered.

Please answer each question by ticking the appropriate box.

A. General Issues

	Yes	No
5.1: Does the study involve participants age 16 or over who are unable to give informed consent? (e.g. people with cognitive impairment, learning disabilities, mental health conditions, physical or sensory impairments?)	☒	☐
5.2: Does the research involve other vulnerable groups such as children (aged under 16) or those in unequal relationships with the researcher? (e.g. your own students)	☒	☐
5.3: Will this research require the cooperation of a gatekeeper* for initial access to the groups or individuals to be recruited?	☒	☐
5.4: Will this research involve discussion of sensitive topics (e.g. sexual activity, drug use, physical or mental health)?	☒	☐
5.5: Could the study induce psychological stress or anxiety or cause harm or negative consequences beyond the risks encountered in normal life?	☒	☐
5.6: Are drugs, placebos or other substances (e.g. food substances, vitamins) to be administered to the study participants or will the study involve invasive, intrusive or potentially harmful procedures of any kind?	☐	☒
5.7: Will this research involve people taking part in the study without their knowledge and consent at the time?	☒	☐
5.8: Does this research involve the internet or other visual/vocal methods where people may be identified?	☒	☐
5.9: Will this research involve access to personal information about identifiable individuals without their knowledge or consent?	☐	☒
5.10: Does the research involve recruiting members of the public as researchers (participant research)?	☐	☒
5.11: Will the research involve administrative or secure data that requires permission from the appropriate authorities before use?	☐	☒

5.12: Is there a possibility that the safety of the researcher may be in question?	☐	☒
5.13: Will financial inducements (other than reasonable expenses and compensation for time) be offered to participants?	☐	☒

*Gatekeeper- a person who controls or facilitates access to the participants

B. Before starting data collection

	Yes	No
6.12: My full identity will be revealed to all research participants.	☒	☐
6.13: All participants will be given accurate information about the nature of the research and the purposes to which the data will be put. (<i>An example of a Participant Information Sheet is available for you to amend and use at xxxxx</i>) http://www.nottingham.edu.cn/en/research/documents/participant-information-sheet-in-english-and-chinese.doc	☒	☐
6.14: All participants will freely consent to take part, and, where appropriate, this will be confirmed by use of a consent form. (An example of a Consent Form is available for you to amend and use at: http://www.nottingham.edu.cn/en/research/researchethics/ethics-approval-process.aspx)	☒	☐
6.15: All participants will freely consent to take part, but due to the qualitative nature of the research a formal consent form is either not feasible or is undesirable and alternative means of recording consent are proposed.	☐	☒
6.16: A signed copy of the consent form or (where appropriate) an alternative record of evidence of consent will be held by the researcher.	☒	☐
6.17: It will be made clear that declining to participate will have no negative consequences for the individual.	☒	☐
6.18: Participants will be asked for permission for quotations (from data) to be used in research outputs where this is intended.	☒	☐
6.19: I will inform participants how long the data collected from them will be kept.	☒	☐
6.20: Incentives (other than basic expenses) will be offered to potential participants as an inducement to participate in the research. (Here any incentives include cash payments and non-cash items such as vouchers and book tokens.)	☐	☒
6.21: For research conducted within, or concerning, organisations (e.g. universities, schools, hospitals, care homes, etc) I will gain authorisation in advance from an appropriate committee or individual.	☒	☐

C. During the process of data collection

	Yes	No
6.25: I will provide participants with my University contact details, and those of my supervisor (<i>where applicable</i>) so that they may get in touch about any aspect of the research if they wish to do so.	☒	☐
6.26: Participants will be guaranteed anonymity only insofar as they do not disclose any illegal activities.	☒	☐



6.27: Anonymity will not be guaranteed where there is disclosure or evidence of significant harm, abuse, neglect or danger to participants or to others.	☒	☐
6.28: All participants will be free to withdraw from the study at any time, including withdrawing data following its collection.	☒	☐
6.29: Data collection will take place only in public and/or professional spaces (e.g. in a work setting)	☒	☐
6.30: Research participants will be informed when observations and/or recording is taking place.	☒	☐
6.31: Participants will be treated with dignity and respect at all times.	☒	☐

D. After collection of data

	Yes	No
6.32: Where anonymity has been agreed with the participant, data will be anonymised as soon as possible after collection.	☒	☐
6.33: All data collected will be stored in accordance with the requirements of the University's Code of Research Conduct	☒	☐
6.34: Data will only be used for the purposes outlined within the participant information sheet and the agreed terms of consent.	☒	☐
6.35: Details which could identify individual participants will not be disclosed to anyone other than the researcher, their supervisor and (if necessary) the Research Ethics Panel and external examiners without participants' explicit consent.	☒	☐

E. After completion of research

	Yes	No
6.37: Participants will be given the opportunity to know about the overall research findings.	☒	☐
6.38: All hard copies of data collection tools and data which enable the identification of individual participants will be destroyed.	☒	☐

If you have **not** ticked any shaded boxes, please send the completed and signed form to the School's Research Ethics Officers, with any further required documents, for approval and record-keeping¹.

If you have ticked any shaded boxes **you will need to describe more fully how you plan to deal with the ethical issues raised by your research.** Issues to consider in preparing an ethics review are given below. Please send this completed form to the Research Ethics Officer who will decide whether your project requires further review by the UNNC Research Ethics Sub-Committee and/or whether further information needs to be provided. Please note that it is your responsibility to follow the University's **Research Code of Conduct** and any relevant academic or professional guidelines in the conduct of your study. **This includes providing appropriate information sheets and**

¹ Investigator's considerations about potential ethical issues in this study is attached at the end of the form. Please kindly see the statements in the last pages.

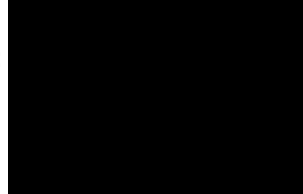
consent forms, and ensuring confidentiality in the storage and use of data. For guidance and UK regulations on the latter, please refer to the Data Protection Policy and Guidelines of the University of Nottingham:

Policy - <http://www.nottingham.ac.uk/%7Ebrzdpa/local/dp-policy.doc>

Guidelines - <http://www.nottingham.ac.uk/~brzdpa/local/dp-guidance.doc>

Any significant change in the project question(s), design or conduct over the course of the research should be notified to the School Research Ethics Officer and may require a new application for ethical approval.

Signature of Principal Investigator/Researcher:
Signature of Supervisor (where appropriate):
Date 15 January 2022



Research Ethics Panel response

- ☒ the research can go ahead as planned
- ☐ further information is needed on the research protocol (see details below)
- ☐ amendments are requested to the research protocol (see details below)

School



A. LIST OF POINTS TO CONSIDER WHEN SUBMITTING AN ETHICS REVIEW (taken from ESRC (2012) *Framework for Research Ethics*).

Risks

1. Have you considered risks to:
 - the research team?
 - the participants? *Eg harm, deception, impact of outcomes*
 - the data collected? *Eg storage, considerations of privacy, quality*
 - the research organisations, project partners and funders involved?
2. Might anyone else be put at risk as a consequence of this research?
3. What might these risks be?
4. How will you protect your data at the research site and away from the research site?
5. How can these risks be addressed?

Details and recruitment of participants

6. What types of people will be recruited? *Eg students, children, people with learning disabilities, elderly?*
7. How will the competence of participants to give informed consent be determined?

8. How, where, and by whom participants will be identified, approached, and recruited?
9. Will any unequal relationships exist between anyone involved in the recruitment and the potential participants?
10. Are there any benefits to participants?
11. Is there a need for participants to be de-briefed? By whom?

Research information

12. What information will participants be given about the research?
13. Who will benefit from this research?
14. Have you considered anonymity and confidentiality?
15. How will you store your collected data?
16. How will data be disposed of and after how long?
17. Are there any conflicts of interest in undertaking this research? *Eg financial reward for outcomes etc.*
18. Will you be collecting information through a third party?

Consent

19. Have you considered consent?
20. If using secondary data, does the consent from the primary data cover further analysis?
21. Can participants opt out?
22. Does your information sheet (or equivalent) contain all the information participants need?
23. If your research changes, how will consent be renegotiated?

Ethical procedures

24. Have you considered ethics within your plans for dissemination/impact?
25. Are there any additional issues that need to be considered? *Eg local customs, local 'gatekeepers', political sensitivities*
26. Have you considered the time you need to gain ethics approval?
27. How will the ethics aspects of the project be monitored throughout its course?
28. Is there an approved research ethics protocol that would be appropriate to use?
29. How will unforeseen or adverse events in the course of research be managed? *Eg do you have procedures to deal with any disclosures from vulnerable participants?*

Ethical Considerations

For this research, there are a range of potential ethical issues: 5.1, 5.2, 5.3, 5.4, 5.5, 5.7 and 5.8, which should be addressed:

The study aims to evaluate the SRMS system targeting children with ASD. This aim requires the study to involve young participants diagnosed as ASD. A control group consisting of TD children will be set up as well. The above recruitment plan relates to the ethical issues 5.1, 5.2 and 5.7, as all the participants are under 16 and some are especially vulnerable as ASD individuals who are below the legal age of consent. ASD children may be unable to recognise this is a testing even though it is explained to them. To address these ethical issues, the investigator has discussed with the supervisors and advisors from collaborative institutes. They are quite supportive of the study and would like to help make the testing friendly to children and their parents. Participation in the study will be voluntary and informed consent will be obtained from their parents or legal guardians in advance.

This study has also gained support from childcare professionals.

[REDACTED]

[REDACTED] will provide support during the whole testing.

Before the testing, the parents or legal guardians will complete the registration on the application, which infers their consent to the study. Children's sensory profile questionnaire answers will be obtained in the registration phase as well. The investigator will assist the parents or legal guardians on how to complete the registration. Parents will give informed consent that they agree to their child's participation and allow the investigator to access the child's sensory profiles.

As mentioned, the testing will be conducted in controlled or uncontrolled settings. In controlled environment, adjustments to variables will not induce psychological stress or anxiety or cause harm or negative consequences beyond the risks encountered in normal life. However, some children with ASD might be hyper-sensitive to external stimuli. Some children are also hyper-sensitive to wearable devices. It is unlikely to exclude any possibility of inducing stress or anxiety in children with ASD as described in 5.5. What the investigator plans to do to minimise the risk is to keep the settings within the safety limits on healthy level. The on-site professionals and caregivers can say stop if they spot distress from the child. A second attempt will be given with caregivers' consent. Children should be withdrawn from the study if they are distressed again in the second attempt. **The parents will be informed that their withdrawal will not affect their children's rehabilitation or education in the future. When any problems in the child are identified during the testing, the on-site professionals will provide timely and effective interventions.**

Previously in 2019, the investigator has obtained UNNC ethical approval (see Figure 4) for a pilot study using the prototype of the SRMS system. The pilot study finally involved 35 children with ASD. In the previous study, the participants showed high acceptance of the prototype and the study procedure. This experience helps the investigator gain the capability of managing ethical issues when operating the testing. Besides, in this study, ASD children will undertake the testing in real-life environment with their teachers, which will increase children's sense of security and lower the risk.

From: Georgios Kapogiannis <Georgios.Kapogiannis@nottingham.edu.cn>

Sent: Monday, June 24, 2019 4:34 PM

Subject: RE: Research Ethics

Importance: High

Dear All,

On behalf of the team I would like to congratulate you! It is one of most complete and comprehensive forms we have seen!

The ethics panel has approved your application. Please feel free to start when you are ready!

Many thanks

On behalf of the team

Georgios

Figure 4: Ethical approval for the previous prototype test

The study will involve using the school monitoring system to record the participants' performance. These video records will only be kept within the collaborative schools in accordance with the school's security law. The investigator may need to access the record in case there happens to be missing data which should have been collected via observation. The access to the video records should be within the collaborative schools on request made by the investigator.

The SRMS system used in this study also involves the internet for sending raw data to the cloud database. To address the issue 5.8, the investigator will anonymise the raw data by suggesting parents not to use children's real names as the account name in the registration phase. Date of birth and gender are the only personal information about children that should be correctly provided. This can prevent privacy-sensitive information to be sent to the cloud database. Data stored on the cloud database are protected by the password which can only be accessed by the investigator. Besides, children's real names will not be shown in any paper documents such as post-session survey forms or consent forms. Their unique account name will be used instead. Paper documents will be stored

securely in locked file cabinets when not in use. SPSS and python will be used for data analysis. Electronic data will be stored on investigator's personal computer with secured access (i.e., automatic lock out feature, password to the computer). All the data will be treated confidentially and that neither parents, nor children's identifiable information will be mentioned in the study or any publications of the results.

Although the research will require the cooperation of 'gatekeepers', e.g., school coordinators, for access to the sites and participants, the role of 'gatekeepers' will be made clear, specifically in advising and as key informants facilitating the access to participants. The on-site teachers will also ensure that ethical obligations are carried out in the school settings.

Working on such a sensitive topic, in addition to its research purposes, the investigator hopes to make the study safe and full of educational purposes for children with and without ASD. This is why the investigator selects classrooms in childcare centres within which the participants will feel safe and comfortable, rather than an office room on campus, as the testing sites. During the whole testing, the investigator will always comply with the University's Code of Research Conduct and Research Ethics.

Contact details of collaborative institutes:

[REDACTED]
[REDACTED]

[REDACTED]
[REDACTED]